

SQL Stored Procedures

22 · Stored Procedures — Complete Reference Guide

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1. Basics — Create and Execute

→ *Define a basic stored procedure*

```
CREATE PROCEDURE GetCustomerSummary AS
BEGIN
    SELECT
        COUNT(*) AS TotalCustomers,
        AVG(Score) AS AvgScore
    FROM Sales.Customers
    WHERE Country = 'USA';
END
GO
```

→ *Execute the stored procedure*

```
EXEC GetCustomerSummary;
```

2. Parameters

→ *ALTER to add a parameter with a default value*

```
ALTER PROCEDURE GetCustomerSummary
    @Country NVARCHAR(50) = 'USA'
AS
```

```

BEGIN
    SELECT
        COUNT(*) AS TotalCustomers,
        AVG(Score) AS AvgScore
    FROM Sales.Customers
    WHERE Country = @Country;
END
GO

```

→ *Execute with different parameter values*

```

EXEC GetCustomerSummary @Country = 'Germany';
EXEC GetCustomerSummary @Country = 'USA';
EXEC GetCustomerSummary; -- uses default 'USA'

```

3. Multiple Queries

→ *Run two independent reports inside one procedure*

```

ALTER PROCEDURE GetCustomerSummary
    @Country NVARCHAR(50) = 'USA'
AS
BEGIN
    -- Query 1: Customer count and average score
    SELECT
        COUNT(*) AS TotalCustomers,
        AVG(Score) AS AvgScore
    FROM Sales.Customers
    WHERE Country = @Country;

    -- Query 2: Order count and total sales
    SELECT
        COUNT(OrderID) AS TotalOrders,
        SUM(Sales) AS TotalSales
    FROM Sales.Orders AS o
    JOIN Sales.Customers AS c
        ON c.CustomerID = o.CustomerID
    WHERE c.Country = @Country;
END
GO

```

4. Variables

→ *Declare variables, capture query results, print to console*

```

ALTER PROCEDURE GetCustomerSummary
    @Country NVARCHAR(50) = 'USA'
AS
BEGIN

```

```

DECLARE @TotalCustomers INT, @AvgScore FLOAT;

SELECT
    @TotalCustomers = COUNT(*),
    @AvgScore       = AVG(Score)
FROM Sales.Customers
WHERE Country = @Country;

PRINT('Total Customers from ' + @Country + ':'
      + CAST(@TotalCustomers AS NVARCHAR));
PRINT('Average Score from '   + @Country + ':'
      + CAST(@AvgScore AS NVARCHAR));

END
GO

```

5. Control Flow — IF/ELSE

→ *Check for NULL scores and update before generating report*

```

ALTER PROCEDURE GetCustomerSummary
    @Country NVARCHAR(50) = 'USA'
AS
BEGIN
    DECLARE @TotalCustomers INT, @AvgScore FLOAT;

    -- Prepare & Cleanup
    IF EXISTS (
        SELECT 1 FROM Sales.Customers
        WHERE Score IS NULL AND Country = @Country
    )
    BEGIN
        PRINT('Updating NULL Scores to 0');
        UPDATE Sales.Customers
        SET Score = 0
        WHERE Score IS NULL AND Country = @Country;
    END
    ELSE
        PRINT('No NULL Scores found');

    -- Report
    SELECT
        @TotalCustomers = COUNT(*),
        @AvgScore       = AVG(Score)
    FROM Sales.Customers
    WHERE Country = @Country;

    PRINT('Total: ' + CAST(@TotalCustomers AS NVARCHAR));
    PRINT('Avg:   ' + CAST(@AvgScore AS NVARCHAR));

```

```
END
GO
```

6. Error Handling — TRY/CATCH

→ *Wrap entire procedure body in TRY/CATCH for robust error handling*

```
ALTER PROCEDURE GetCustomerSummary
    @Country NVARCHAR(50) = 'USA'
AS
BEGIN
    BEGIN TRY
        DECLARE @TotalCustomers INT, @AvgScore FLOAT;

        SELECT
            @TotalCustomers = COUNT(*),
            @AvgScore        = AVG(Score)
        FROM Sales.Customers
        WHERE Country = @Country;

        PRINT('Total: ' + CAST(@TotalCustomers AS NVARCHAR));
        PRINT('Avg:    ' + CAST(@AvgScore AS NVARCHAR));

        SELECT
            COUNT(OrderID) AS TotalOrders,
            SUM(Sales)      AS TotalSales,
            1/0             AS FaultyCalculation -- intentional error
        FROM Sales.Orders AS o
        JOIN Sales.Customers AS c
            ON c.CustomerID = o.CustomerID
        WHERE c.Country = @Country;
    END TRY
    BEGIN CATCH
        PRINT('An error occurred.');
```

```
        PRINT('Error Message:  ' + ERROR_MESSAGE());
        PRINT('Error Number:    ' + CAST(ERROR_NUMBER() AS NVARCHAR));
        PRINT('Error Severity:  ' + CAST(ERROR_SEVERITY() AS NVARCHAR));
        PRINT('Error State:     ' + CAST(ERROR_STATE() AS NVARCHAR));
        PRINT('Error Line:      ' + CAST(ERROR_LINE() AS NVARCHAR));
        PRINT('Error Procedure: ' + ISNULL(ERROR_PROCEDURE(), 'N/A'));
    END CATCH;
END
GO
```

→ *Execute the final stored procedure*

```
EXEC GetCustomerSummary @Country = 'Germany';
EXEC GetCustomerSummary @Country = 'USA';
EXEC GetCustomerSummary;
```

